

An Arbitrary ℓ_1 -Sum Extension for Property $CWO-S$

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Status

This packet gives a likely valid partial result for the natural question of Haller–Kuuseok–Poldvere [1]. The source paper proves that $L_1(\mu, X)$ has property $CWO-S$ whenever μ is non-zero σ -finite and X is weakly uniformly rotund, and asks whether “weakly uniformly rotund” can be replaced by “has property $CWO-S$ ”.

The result here proves the positive answer on the purely atomic side in a slightly stronger form: arbitrary, not merely countable, ℓ_1 -sums of $CWO-S$ spaces have $CWO-S$. Consequently $L_1(\mu, X)$ has $CWO-S$ for arbitrary purely atomic measure spaces whenever X has $CWO-S$. The non-atomic case, and the injective tensor-product question, remain open here.

Source Problem Evidence

The source question appears in the introduction of [1], page 3 of the arXiv PDF:

Then also the 1-sum $\bigoplus_{i=1}^{\infty} X_i$ has property $CWO-S$.

Finally, in Section 4, we prove Theorem 1.1.

A natural question arises in connection with Theorem 1.1: does this theorem remain true when X is assumed to be any Banach space with property $CWO-S$ (note that strictly convex spaces—and thus, as a special case, wUR spaces—do have property $CWO-S$) or, moreover, does the injective tensor product of two Banach spaces with property $CWO-S$ have property $CWO-S$?

2. AUXILIARY RESULTS FOR THE PROOF OF THEOREM 1.1

In [2, Theorem 5.5] it was proven that, for any non-zero σ -finite (countably additive non-negative) measure μ , the real space $L_1(\mu)$ has property $CWO-S$. However, the proof in [2] contains a gap: it is valid only for the case when, using the no-

Transcription

After proving the wUR -valued L_1 -theorem and the countable ℓ_1 -sum stability proposition, the authors ask whether the theorem remains true when X is assumed to be any Banach space with property $CWO-S$, and moreover whether the injective tensor product of two Banach spaces with property $CWO-S$ must have property $CWO-S$.

Definitions

A Banach space X has property *CWO-S* if, for every finite convex combination

$$C = \sum_{k=1}^n \lambda_k U_k$$

of relatively weakly open subsets U_k of B_X , every point of $C \cap S_X$ is an interior point of C in the relative weak topology of B_X .

For a family $(X_i)_{i \in I}$, write

$$\left(\bigoplus_{i \in I} X_i \right)_1$$

for the Banach space of all families $x = (x_i)$ with $\sum_{i \in I} \|x_i\| < \infty$, normed by that sum.

Proof Intuition

The source paper proves countable ℓ_1 -sum stability. In an arbitrary ℓ_1 -sum, any two fixed vectors x, y have countable support, so their norm-one convex combination $z = \lambda x + \nu y$ lives in a countable one-complemented summand A . That summand has *CWO-S* by the source proposition. A weakly small perturbation w of z in the whole arbitrary sum may have a small component b outside A . Decompose the normalized A -part of w inside A , then put the same off-support component b into both summands. The ℓ_1 -norm budget absorbs this extra component exactly because $\|P_A w\| + \|(I - P_A)w\| = \|w\| \leq 1$.

Theorem 1. *Let $(X_i)_{i \in I}$ be an arbitrary family of Banach spaces with property *CWO-S*. Then*

$$Z = \left(\bigoplus_{i \in I} X_i \right)_1$$

*has property *CWO-S*.*

Proof. We use the standard reduction to convex combinations of two relatively weakly open sets, as in Lemma 4.1(b) of [2] and in the proof of Proposition 1.2 of [1]. Thus let U, V be relatively weakly open subsets of B_Z , let $\lambda, \nu > 0$ with $\lambda + \nu = 1$, and suppose

$$x \in U, \quad y \in V, \quad z = \lambda x + \nu y \in S_Z.$$

It is enough to find a relative weak neighbourhood W of z in B_Z such that $W \subset \lambda U + \nu V$.

Choose a finite set $F \subset S_{Z^*}$ and $\eta > 0$ such that

$$U_0 = \{u \in B_Z : |f(u - x)| < 3\eta \text{ for all } f \in F\} \subset U$$

and similarly

$$V_0 = \{v \in B_Z : |f(v - y)| < 3\eta \text{ for all } f \in F\} \subset V.$$

Let $J = \text{supp } x \cup \text{supp } y$. This set is countable. Put

$$A = \left(\bigoplus_{i \in J} X_i \right)_1, \quad B = \left(\bigoplus_{i \in I \setminus J} X_i \right)_1$$

so that $Z = A \oplus_1 B$, and let $P : Z \rightarrow A$ be the norm-one coordinate projection. The vectors x, y, z belong to A . By Proposition 1.2 of [1], A has property $CWO-S$.

For $f \in F$, denote by f_A and f_B its restrictions to A and B . Define relatively weakly open neighbourhoods of x and y in B_A by

$$U_A = \{a \in B_A : |f_A(a - x)| < \eta \text{ for all } f \in F\},$$

$$V_A = \{a \in B_A : |f_A(a - y)| < \eta \text{ for all } f \in F\}.$$

Since A has $CWO-S$ and $z = \lambda x + \nu y \in S_A$, there is a relative weak neighbourhood W_A of z in B_A such that

$$W_A \subset \lambda U_A + \nu V_A.$$

Choose a finite set $G \subset A^*$ and $\delta > 0$ such that

$$\{a \in B_A : |g(a - z)| < \delta \text{ for all } g \in G\} \subset W_A.$$

Let $h \in S_{A^*}$ satisfy $h(z) = 1$. Decreasing $\rho > 0$ if necessary, assume

$$\rho < \eta, \quad \rho < \delta/4, \quad \rho(1 + \|g\|) < \delta/2 \quad (g \in G).$$

Now set

$$W = \{w \in B_Z : |g(Pw - z)| < \rho \ (g \in G), \quad |h(Pw - z)| < \rho\}.$$

This is a relative weak neighbourhood of z in B_Z .

Fix $w \in W$. Write $a = Pw \in A$ and $b = w - a \in B$, and set $t = \|a\|$. Since $h(a) > 1 - \rho$, we have $t > 1 - \rho > 0$, and $\|b\| \leq 1 - t < \rho$. For $g \in G$,

$$\left| g\left(\frac{a}{t} - z\right) \right| \leq \frac{|g(a - z)|}{t} + \left| \frac{1}{t} - 1 \right| |g(z)| < 2\rho + 2\rho\|g\| < \delta.$$

Thus $a/t \in W_A$. Hence there exist $a_1 \in U_A$ and $a_2 \in V_A$ such that

$$\frac{a}{t} = \lambda a_1 + \nu a_2.$$

Define

$$u = ta_1 + b, \quad v = ta_2 + b$$

as elements of $A \oplus_1 B = Z$. Then

$$\lambda u + \nu v = t(\lambda a_1 + \nu a_2) + b = a + b = w.$$

Also

$$\|u\| \leq t\|a_1\| + \|b\| \leq t + \|b\| = \|w\| \leq 1,$$

and similarly $\|v\| \leq 1$.

It remains to check $u \in U_0$ and $v \in V_0$. For $f \in F$,

$$f(u - x) = tf_A(a_1 - x) + (t - 1)f_A(x) + f_B(b).$$

Since $a_1 \in U_A$, $\|f_A\| \leq 1$, $\|x\| \leq 1$, $|1 - t| < \rho$, and $\|b\| < \rho$, we get

$$|f(u - x)| < \eta + \rho + \rho < 3\eta.$$

Thus $u \in U_0 \subset U$. The same estimate with a_2, y, V_A gives $v \in V_0 \subset V$. Therefore $w \in \lambda U + \nu V$. Since $w \in W$ was arbitrary, $W \subset \lambda U + \nu V$, completing the proof. $\square$

Corollary 1. *Let μ be a purely atomic measure whose finite-measure atoms generate the L_1 -space, and let X be a Banach space with property $CWO-S$. Then $L_1(\mu, X)$ has property $CWO-S$.*

Proof. On each finite-measure atom A , every Bochner integrable function is almost everywhere constant. Therefore $L_1(\mu, X)$ is isometric to the ℓ_1 -sum over the finite-measure atoms, with each summand isometric to X after the harmless scaling by $\mu(A)$. Apply Theorem 1. $\square$

Verification Notes

No computation is needed. The proof uses only:

- the source paper’s countable ℓ_1 -sum stability theorem;
- the standard two-set reduction for property $CWO-S$;
- the elementary fact that every element of an arbitrary ℓ_1 -sum has countable support;
- the norm identity in $A \oplus_1 B$.

Limitations

This is a partial result. For the source paper’s σ -finite setting, the purely atomic case is already morally countable and close to Proposition 1.2 of [1]. The new contribution is the arbitrary-index ℓ_1 -sum argument and the clean atomic formulation without a countability restriction. The non-atomic $L_1(\mu, X)$ case for general $CWO-S$ spaces and the injective tensor-product question are not settled by this packet.

Novelty and Literature Check

I searched the run indexes for arXiv:2503.09182, $CWO-S$, ℓ_1 -sum, injective tensor product, and convex combinations of relatively weakly open sets. The only overlapping run packet was a prior partial result for the older CWO tensor question from arXiv:1806.10693; it does not contain the arbitrary ℓ_1 -sum upgrade proved here.

I also ran bounded web searches for “arbitrary 1-sum property $CWO-S$ ”, “ ℓ_1 -sum $CWO-S$ Banach spaces”, and related direct-sum phrases. The searches found the source paper’s countable result and adjacent CWO literature, but no exact arbitrary-index statement. Novelty confidence is moderate: the proof is short and natural from the source proposition, so it may be folklore.

Human Review Recommendation

Check the passage from the countable support summand A back to the full sum $A \oplus_1 B$, especially the normalization $a/\|a\| \in W_A$ and the estimate showing $u = ta_1 + b$ and $v = ta_2 + b$ stay in the original weak neighbourhoods. This is the only genuinely new step beyond the source paper’s countable theorem.

References

- [1] R. Haller, P. Kuuseok, and M. Poldvere, *Norm-one points in convex combinations of relatively weakly open subsets of the unit ball in the spaces $L_1(\mu, X)$* , arXiv:2503.09182, 2025. <https://arxiv.org/abs/2503.09182>
- [2] T. A. Abrahamsen, J. Becerra Guerrero, R. Haller, V. Lima, and M. Poldvere, *Banach spaces where convex combinations of relatively weakly open subsets of the unit ball are relatively weakly open*, *Studia Math.* **250** (2020), no. 3, 297–320. <https://arxiv.org/abs/1806.10693>